SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.*

QUESTIONS: 1 **Total Marks:100**

Annexure A

Technical Presentation

Code of Conduct:

I Akshaya Rajesh (Name 192 24086) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Akshaya A.R.
Signature of the student:

95/100

Annexure A

Technical Presentation [Any one]

Question 1
Suppose an Operating System is used in a banking system handling thousands of transactions simultaneously.
Prepare a technical presentation explaining:
Types of OS suitable for this scenario
OS structure used
Role of system calls in transaction processing **(BL3)**

Question 2
Consider a multi-user Linux-based server environment supporting file sharing and device access.
Prepare a technical presentation covering:
OS functions involved in resource management
File and device system calls
Interaction between user processes and kernel **(BL3)**

Question 3
Assume a real-time embedded system used in traffic signal control.
Prepare a technical presentation explaining:
Type of OS used and justification
OS structure and components
System calls used for process control and timing **(BL3)**

Question 4
A company uses a distributed computing system across multiple locations.
Prepare a technical presentation discussing:
Type of OS and its characteristics
Architecture of distributed OS
System calls used for communication and coordination **(BL3)**

Question 5
Consider an OS managing process scheduling, memory allocation, and I/O operations in a modern computer.
Prepare a technical presentation including:
Key OS functions and their importance
Interaction between OS components
Examples of system calls for each function **(BL3)**

RUBRICS FOR CALCULATION

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Operating Systems in Banking Systems

TECHNICAL PRESENTATION

PRESENTED BY
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Introduction

- Banking systems handle thousands of transactions simultaneously.
- An Operating System (OS) manages hardware, memory, processes, and system resources.
- It ensures efficient, secure, and reliable transaction processing.
- It supports services such as ATM operations, online banking, and money transfers.

Types of OS Suitable for Banking

- Multi-user OS: Supports multiple users accessing banking services simultaneously.
- Multiprocessing OS: Uses multiple CPU cores to process thousands of transactions efficiently.
- Real-time OS features: Supports time-sensitive operations such as payment authorization and fraud alerts.
- **Example**: Enterprise Linux or UNIX server operating systems.

Role of System Calls in Transaction Processing

- Process Control: Manages banking service processes.
- File Management: Reads and writes transaction logs.
- Networking: Communicates with ATMs and payment gateways.
- Synchronization: Coordinates concurrent operations.
- Security: Controls access to system resources.

Transaction Workflow

Example: Online Money Transfer

- Request: Customer initiates a transfer.
- Validation: System verifies identity and account balance.
- Processing: Banking application requests account updates.
- Commit: Database transaction records the changes reliably.
- Confirmation: Customer receives the transaction status.


Thank
you!